

DONGTAI, China

WMO: 582510

Lat: 32.8576N

Lon: 120.2765E

Elev: 4

StdP: 101.28

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.7	-3.4	-12.7	1.3	-0.1	-10.4	1.5	-0.2	8.8	1.6	7.8	2.5	2.1	290	0.407

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.3	34.6	27.6	33.4	27.1	32.2	26.6	28.7	32.7	28.1	32.1	27.5	31.2	3.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.7	23.7	31.0	27.1	22.8	30.5	26.5	22.0	30.0	93.4	32.7	90.6	32.2	87.8	31.4	31.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.6	6.5	5.7	-6.7	36.5	1.5	1.1	-7.8	37.2	-8.6	37.8	-9.5	38.4	-10.5	39.2
			-7.8	29.7	1.7	0.9	-9.1	30.3	-10.1	30.8	-11.1	31.2	-12.4	31.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	15.8	2.8	4.6	9.1	14.8	20.2	24.2	28.0	27.5	23.2	17.9	11.5	5.2	(6)
(7)		DBStd	9.25	3.15	3.46	3.66	3.57	3.00	2.55	2.68	2.45	2.54	2.90	3.79	3.47	(7)
(8)		HDD10.0	626	225	153	62	4	0	0	0	0	0	0	27	154	(8)
(9)		HDD18.3	1941	483	384	288	116	14	0	0	0	1	43	205	408	(9)
(10)		CDD10.0	2747	1	3	33	148	317	427	557	543	397	246	73	4	(10)
(11)		CDD18.3	1021	0	0	0	9	73	177	298	284	147	30	1	0	(11)
(12)		CDH23.3	9327	0	0	4	83	527	1313	3365	3028	904	99	3	0	(12)
(13)		CDH26.7	3476	0	0	0	14	153	439	1483	1177	201	9	0	0	(13)
(14)	Wind	WSAvg	2.7	2.7	2.9	3.2	3.1	2.9	2.8	2.7	2.8	2.5	2.3	2.5	2.6	(14)
(15)	Precipitation	PrecAvg	1036	27	41	54	75	84	139	221	169	121	54	54	22	(15)
(16)		PrecMax	1428	66	100	111	197	217	350	720	534	382	177	202	75	(16)
(17)		PrecMin	462	0	0	13	19	19	42	25	6	20	2	0	0	(17)
(18)		PrecStd	208	15	28	26	46	42	88	150	114	84	45	50	20	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.9	17.6	23.4	28.7	32.9	34.3	36.2	35.6	32.7	27.9	23.2	16.5	(19)
(20)			MCWB	9.8	12.4	16.2	20.9	21.2	25.2	28.3	28.4	26.1	21.4	16.9	12.4	(20)
(21)		2%	DB	11.0	14.3	20.1	25.6	30.3	32.4	35.0	34.1	30.4	25.7	20.7	13.8	(21)
(22)			MCWB	7.8	9.7	14.1	18.7	20.8	24.1	28.1	27.9	25.1	19.3	15.9	10.4	(22)
(23)		5%	DB	9.1	12.0	17.5	23.5	28.2	30.8	33.9	32.9	28.8	24.3	18.9	11.9	(23)
(24)			MCWB	6.4	8.4	12.4	17.4	19.8	23.4	27.8	27.5	24.0	18.8	15.1	9.1	(24)
(25)		10%	DB	7.6	10.1	15.3	21.3	26.3	29.3	32.7	31.7	27.5	22.8	17.4	10.4	(25)
(26)			MCWB	5.5	7.3	11.3	16.0	19.1	23.0	27.4	27.0	23.2	18.2	14.2	8.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.4	13.9	17.3	22.3	24.6	27.5	29.6	29.5	28.0	23.4	19.5	14.6	(27)
(28)			MCD B	12.8	17.0	21.7	26.6	28.6	31.9	33.4	33.4	31.7	25.2	20.7	15.3	(28)
(29)		2%	WB	8.6	10.9	15.1	20.3	22.8	26.5	28.9	28.6	26.6	21.7	17.6	11.3	(29)
(30)			MCD B	10.1	13.5	18.3	24.0	26.6	30.3	33.1	32.4	29.0	23.8	19.2	12.9	(30)
(31)		5%	WB	7.1	9.3	13.5	18.6	21.6	25.6	28.3	28.0	25.6	20.5	16.0	9.9	(31)
(32)			MCD B	8.5	11.1	16.9	22.1	25.4	28.7	32.5	31.7	27.5	22.6	18.1	11.3	(32)
(33)		10%	WB	5.8	7.7	11.8	16.9	20.5	24.5	27.8	27.5	24.5	19.3	14.5	8.5	(33)
(34)			MCD B	7.3	9.6	14.5	20.2	24.2	27.2	31.8	30.8	26.4	21.7	16.6	10.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.8	7.3	8.5	9.2	7.5	6.3	6.0	7.0	8.1	8.0	7.3	(35)	
(36)			MCD BR	9.3	10.8	12.4	12.7	12.5	10.1	7.9	7.5	7.9	9.5	9.8	9.3	(36)
(37)			MCWBR	6.1	6.8	7.2	6.8	5.1	3.6	2.9	2.9	3.2	3.9	5.1	5.9	(37)
(38)		5% WB	MCD BR	7.6	8.9	11.0	11.0	9.9	7.6	7.6	7.2	6.4	7.1	7.5	7.3	(38)
(39)			MCWBR	5.8	6.4	7.4	7.1	5.4	3.7	3.3	3.0	3.3	3.9	5.0	5.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.575	0.631	0.656	0.660	0.656	0.745	0.743	0.691	0.615	0.632	0.624	0.577	(40)	
(41)		taud	1.671	1.595	1.567	1.610	1.653	1.531	1.579	1.660	1.770	1.657	1.633	1.681	(41)	
(42)		Ebn at Noon	597	616	648	675	684	624	622	645	673	606	551	564	(42)	
(43)		Edh at Noon	200	238	262	261	252	284	270	245	211	220	206	189	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.33	2.74	3.85	4.59	4.84	4.42	4.65	4.55	3.90	3.28	2.47	2.21	(44)	
(45)		RadStd	0.28	0.46	0.37	0.51	0.46	0.54	0.60	0.66	0.36	0.44	0.29	0.28	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page